Learning Objectives

Learners will be able to...

- Describe how a chatbot works
- Create a simple chatbot in Python with a tkinter GUI
- Create a simple chatbot using NLTK with a tkinter GUI

info

Make Sure to Know

Intermediate Python.

Limitations

In this assignment, we only work with the tkinter and NLTK libraries.

Introduction to Chatbots

One of the most common tasks and a subset of Natural Language Processing is **Natural Language Generation** (NLG).

- It is also sometimes described as the opposite of speech recognition.
- It's the task of generating human language text response by putting together structured information.
- This information follows **syntactical** and **semantic** rules from a program's knowledge base.

The best examples of real world applications of this concept are **virtual agents** and **chatbots**.
- Apple's Siri and Amazon's Alexa are examples of virtual agents.
- They use:
- **Speech recognition** to understand user commands
- **NLG** to respond with the required information or conduct the requested action.

A **Chatbot**, on the other hand, operates on **typed text**.
- It processes and simulates **human dialogs** and **conversations**, allowing users to interact with various kinds of digital services.

- They can be as simple as a program comprised of a set of **hard-coded rules**, to very sophisticated **AI based** chatbots that deliver **personalized responses** and learn as they go.

Connecting to Github

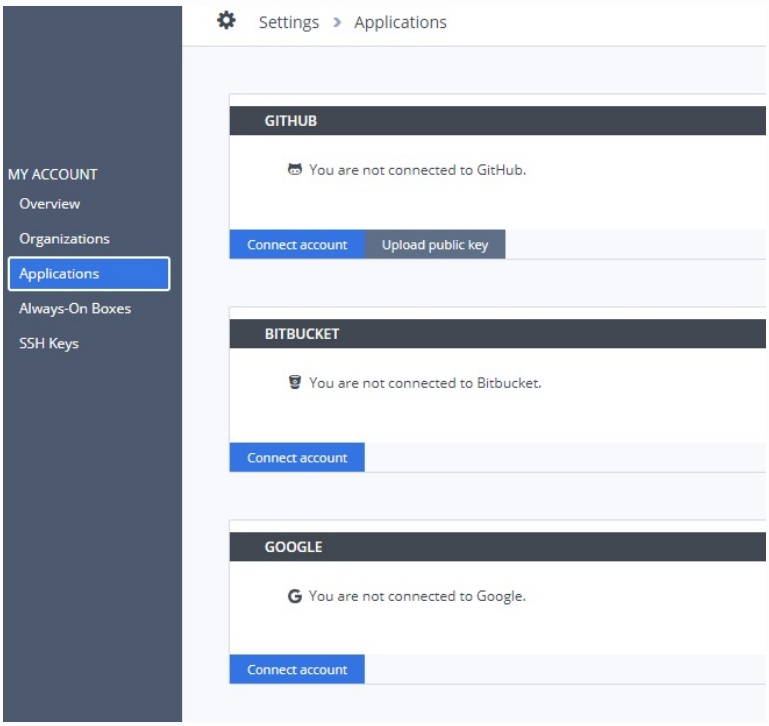
Connecting to GitHub

In this lab, you are going to build a simple Chatbot. In order to showcase this project for your portfolio, you are going to use GitHub. If you do not yet have an account, please create one now. We are going to clone a repo that will contain the code for your Chatbot.

Connecting GitHub and Codio

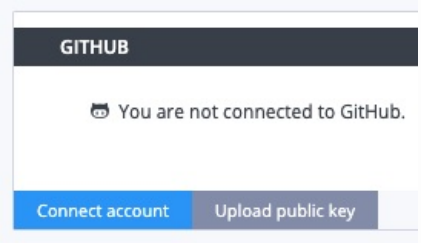
You need to connect GitHub to your Codio account. This only needs to be done one time.

- In your Codio account, click on your username
- Click on **Applications**



Codio Account

- Under GitHub, click on **Connect account**



connect account

- You will be using an SSH connection, so you need to click on **Upload public key**

Fork the Repo

- Go to the nlp-simplebot repo. This repo is the starting point for your project.
- Click on the "Fork" button in the top-right corner.
- Click the green "Code" button.
- Copy the **SSH** information. It should look something like this:

```
git@github.com:<your_github_username>/nlp-simplebot.git
```

info

Important

If you do not use the SSH information, you will have to provide your username and Personal Access Token (PAT) to GitHub each time you push or pull from the repository. See this documentation for setting up a PAT for your GitHub account.

In the Terminal

- Clone the repo. Your command should look something like this:

```
git clone git@github.com:<your_github_username>/nlp-
        simplebot.git
```

- You should see a `nlp-simplebot` directory appear in the file tree.

On the last page, we'll show you how to push to Github.

You are now ready for the lab!

Initializing the GUI

A simple, rule-based chatbot.

info

Writing Code

To your left, you should see an opened file called `simplebot.py`
Use this file and reference the syntax below to work on this Lab.

Let's start by building a GUI for our chatbot. We'll use ***tkinter***, the standard Python GUI library.

The first step is to import all the necessary modules into our file.
Paste the following code in the top left pane:

```
import tkinter.scrolledtext as tks #creates a scrollable text
        window


from datetime import datetime
from tkinter import *
```

To make a simple GUI for our chatbot, it should have 4 main components:
1. `baseWindow` - the main GUI window that contains everything
2. `chatWindow` - displays the conversation between a user and the chatbot
3. `userEntryBox` - for the user to type their queries for the Chatbot
4. `sendButton` - a button that sends the user query to the Chatbot

Let's use `tkinter` to initialize the `baseWindow` and the other components and place them on the `baseWindow`.

Paste the following code in the top left pane under the import statements:

```
baseWindow = Tk()
baseWindow.title("The Simple Bot")
baseWindow.geometry("500x250")

chatWindow = tks.ScrolledText(baseWindow, font="Arial")
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')
chatWindow.config(state=DISABLED)

sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='#ffffff',
    # command=send
)

userEntryBox = Text(baseWindow, bd=1, bg="white", width=38,
        font="Arial")

chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
sendButton.place(x=430, y=200)

baseWindow.mainloop()
```

Let’s close the GUI and look at how we can attach a function to the send button!

Implementing functions for the GUI

At this point, the `sendButton` doesn't do anything.
Let's define a function called `send` which should do the following:

- Collects the `user_input` from the `textBox`
- Gets the `bot_response`
- Inserts both of these into the `chatWindow`

Paste the following code in the top left pane, right under the import statements:

```
def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0",'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0","end")
    chatWindow.see('end')
```

Once we have the `user_input` and `bot_response`, we'll write a couple functions to insert them into the `chatWindow`

Paste the following code in the top left pane, right above the `send` function:

```
def create_and_insert_user_frame(user_input):
  userFrame = Frame(chatWindow, bg="#d0ffff")
  Label(
      userFrame,
      text=user_input,
      font=("Arial", 11),
      bg="#d0ffff").grid(row=0, column=0, sticky="w", padx=5,
        pady=5)
  Label(
      userFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#d0ffff"
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-right')
  chatWindow.window_create('end', window=userFrame)


def create_and_insert_bot_frame(bot_response):
  botFrame = Frame(chatWindow, bg="#ffffd0")
  Label(
      botFrame,
      text=bot_response,
      font=("Arial", 11),
      bg="#ffffd0",
      wraplength=400,
      justify='left'
  ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
  Label(
      botFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#ffffd0"
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-left')
  chatWindow.window_create('end', window=botFrame)
  chatWindow.insert(END, "\n\n" + "")
```

Now, we want to bind the `send` function to the `sendButton`. We can also bind the `Enter` key to the window so we don't have to click on the button everytime.

Replace the `sendButton` initialization in the top left pane with the following code:

```
sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='#ffffff',
    command=send)
sendButton.bind("<Button-1>", send)
baseWindow.bind('<Return>', send)
```

▼ Code

Your code should look like this:

```
import tkinter.scrolledtext as tks #creates a scrollable text
        window

from datetime import datetime
from tkinter import *


def create_and_insert_user_frame(user_input):
  userFrame = Frame(chatWindow, bg="#d0ffff")
  Label(
      userFrame,
      text=user_input,
      font=("Arial", 11),
      bg="#d0ffff").grid(row=0, column=0, sticky="w", padx=5,
        pady=5)
  Label(
      userFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#d0ffff"
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-right')
  chatWindow.window_create('end', window=userFrame)


def create_and_insert_bot_frame(bot_response):
  botFrame = Frame(chatWindow, bg="#ffffd0")
  Label(
      botFrame,
      text=bot_response,
      font=("Arial", 11),
```

```
        bg="#ffffd0",
        wraplength=400,
        justify='left'
    ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
    Label(
        botFrame,
        text=datetime.now().strftime("%H:%M"),
        font=("Arial", 7),
        bg="#ffffd0"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-left')
    chatWindow.window_create('end', window=botFrame)
    chatWindow.insert(END, "\n\n" + "")


def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0",'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0","end")
    chatWindow.see('end')


baseWindow = Tk()
baseWindow.title("The Simple Bot")
baseWindow.geometry("500x250")

chatWindow = tks.ScrolledText(baseWindow, font="Arial")
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')
chatWindow.config(state=DISABLED)

sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='#ffffff',
    command=send)
```

```
sendButton.bind("<Button-1>", send)
baseWindow.bind('<Return>', send)

userEntryBox = Text(baseWindow, bd=1, bg="white", width=38,
        font="Arial")

chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
sendButton.place(x=430, y=200)

baseWindow.mainloop()
```

Now, the only thing left is to define the `bot_response` function!

The bot response function

The `bot_response` function is the logic function for the chatbot. This is where we will define the rules that the chatbot will follow to respond to `user_input`

Paste the following code in the top left pane, right below the import statements:

```
# Generating response
def get_bot_response(user_input):

  bot_response = ""
  if(user_input == "hello"):
    bot_response = "Hi!"
  elif(user_input == "hi" or user_input == "hii" or user_input
       == "hiiii"):
    bot_response = "Hello there! How are you?"
  elif(user_input == "how are you"):
    bot_response = "Oh, I'm great! How about you?"
  elif(user_input == "fine" or user_input == "i am good" or
       user_input == "i am doing good"):
    bot_response = "That's excellent! How can I help you today?"
  else:
    bot_response = "I'm sorry, I don't understand..."

  return bot_response
```

▼ Code

Your code should look like this:

```
import tkinter.scrolledtext as tks #creates a scrollable text
        window

from datetime import datetime
from tkinter import *


# Generating response
def get_bot_response(user_input):

  bot_response = ""
  if(user_input == "hello"):
```

```
    bot_response = "Hi!"
  elif(user_input == "hi" or user_input == "hii" or user_input
        == "hiiii"):
    bot_response = "Hello there! How are you?"
  elif(user_input == "how are you"):
    bot_response = "Oh, I'm great! How about you?"
  elif(user_input == "fine" or user_input == "i am good" or
        user_input == "i am doing good"):
    bot_response = "That's excellent! How can I help you today?"
  else:
    bot_response = "I'm sorry, I don't understand..."

  return bot_response


def create_and_insert_user_frame(user_input):
  userFrame = Frame(chatWindow, bg="#d0ffff")
  Label(
      userFrame,
      text=user_input,
      font=("Arial", 11),
      bg="#d0ffff").grid(row=0, column=0, sticky="w", padx=5,
        pady=5)
  Label(
      userFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#d0ffff"
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-right')
  chatWindow.window_create('end', window=userFrame)


def create_and_insert_bot_frame(bot_response):
  botFrame = Frame(chatWindow, bg="#ffffd0")
  Label(
      botFrame,
      text=bot_response,
      font=("Arial", 11),
      bg="#ffffd0",
      wraplength=400,
      justify='left'
  ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
  Label(
      botFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#ffffd0"
```

```
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-left')
  chatWindow.window_create('end', window=botFrame)
  chatWindow.insert(END, "\n\n" + "")


def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0",'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0","end")
    chatWindow.see('end')


baseWindow = Tk()
baseWindow.title("The Simple Bot")
baseWindow.geometry("500x250")

chatWindow = tks.ScrolledText(baseWindow, font="Arial")
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')
chatWindow.config(state=DISABLED)

sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='#ffffff',
    command=send)
sendButton.bind("<Button-1>", send)
baseWindow.bind('<Return>', send)

userEntryBox = Text(baseWindow, bd=1, bg="white", width=38,
        font="Arial")

chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
sendButton.place(x=430, y=200)
```

```
baseWindow.mainloop()
```

This was the last step! We have successfully built our first chatbot! Let's run it and interact with it!

> important
>
> ## Limitations of this chatbot
>
> This was a simple example of a hard coded chatbot. This doesn't use any NLP concepts and does simple string matching to generate responses.

Let's see how we can make this better by using the `nltk` library

The bot response function using nltk.chat.util

At this point, you must be familiar with NLTK or the Natural Language Toolkit. Along with a suite of text processing libraries, it has an additional package called `nltk.chat`

`nltk.chat` is a class for simple chatbots. It has a built-in module called `nltk.chat.util` that performs simple regex pattern matching on user input and accepts a list of responses for each pattern. Whenever it detects a match, it responds by choosing a response at random from it's respective list.

Let's see how we can use it in our simple bot!

info

Writing Code

To your left, you should see an opened file called `simplebot_nltk.py` Use this file and reference the syntax below to work on this Lab.

In the top left panel, the code file should have the same code as `simplebot.py`, with the exception of an empty `get_bot_response` function.

Replace the `get_bot_response` function by pasting the following code in the top left panel

```
from nltk.chat.util import Chat


# Generating response
def get_bot_response(user_input):
  pairs = [
    ('my name is (.*)', ['Hello ! % 1']),
    ('(hi|hello|hey|holla|hola)', ['Hey there !', 'Hi there !',
        'Hey !']),
    ('(.*) your name ?', ['My name is Charlie', 'I\'m Charlie
        the Chatbot']),
    ('(.*) do you do ?', ['I specialize in regex pattern
        matching! How about you?']),
    ('(.*) created you ?', ['You did! Using python, NLTK and
        tkinter! '])
  ]

  chat = Chat(pairs)
  while user_input[-1] in "!.":
    user_input = user_input[:-1]
  bot_response = chat.respond(user_input)
  return bot_response
```

The `Chat` class from `nltk.chat.util` initializes a chatbot. It takes a parameter called `pairs`, which is a list of tuples, where each tuple is: (pattern, list of responses)

Once we have an instance of the `Chat` class, it provides a method called `respond`, that takes care of the regex pattern matching in the background, and returns the appropriate bot response.

The `Chat` class also has a method called `converse()` that does everything we're doing to clean the punctuation and calls the `respond` function. We're not using `converse` to be able to use `nltk` with a GUI.

▼ Code

Your code should look like this:

```
import tkinter.scrolledtext as tks #creates a scrollable text
        window

from datetime import datetime
from tkinter import *
from nltk.chat.util import Chat


# Generating response
def get_bot_response(user_input):
  pairs = [
```

```
    ('my name is (.*)', ['Hello ! % 1']),
    ('(hi|hello|hey|holla|hola)', ['Hey there !', 'Hi there !',
        'Hey !']),
    ('(.*) your name ?', ['My name is Geeky']),
    ('(.*) do you do ?', ['We provide a platform for tech
        enthusiasts, a wide range of options !']),
    ('(.*) created you ?', ['Geeksforgeeks created me using
        python and NLTK'])
  ]

  chat = Chat(pairs)
  while user_input[-1] in "!.":
    user_input = user_input[:-1]
  bot_response = chat.respond(user_input)
  return bot_response


def create_and_insert_user_frame(user_input):
  userFrame = Frame(chatWindow, bg="#d0ffff")
  Label(
      userFrame,
      text=user_input,
      font=("Arial", 11),
      bg="#d0ffff").grid(row=0, column=0, sticky="w", padx=5,
        pady=5)
  Label(
      userFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
      bg="#d0ffff"
  ).grid(row=1, column=0, sticky="w")

  chatWindow.insert('end', '\n ', 'tag-right')
  chatWindow.window_create('end', window=userFrame)


def create_and_insert_bot_frame(bot_response):
  botFrame = Frame(chatWindow, bg="#ffffd0")
  Label(
      botFrame,
      text=bot_response,
      font=("Arial", 11),
      bg="#ffffd0",
      wraplength=400,
      justify='left'
  ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
  Label(
      botFrame,
      text=datetime.now().strftime("%H:%M"),
      font=("Arial", 7),
```

```
        bg="#ffffd0"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-left')
    chatWindow.window_create('end', window=botFrame)
    chatWindow.insert(END, "\n\n" + "")


def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0",'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0","end")
    chatWindow.see('end')


baseWindow = Tk()
baseWindow.title("The Simple Bot")
baseWindow.geometry("500x250")

chatWindow = tks.ScrolledText(baseWindow, font="Arial")
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')
chatWindow.config(state=DISABLED)

sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='#ffffff',
    command=send)
sendButton.bind("<Button-1>", send)
baseWindow.bind('<Return>', send)

userEntryBox = Text(baseWindow, bd=1, bg="white", width=38,
        font="Arial")

chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
```

```
sendButton.place(x=430, y=200)

baseWindow.mainloop()
```

And that's it! We've built our first chatbot using NLTK! Let's run it and interact with it!

challenge

Try this variation:

- Add more regex patterns and responses to the pairs list to expand on what our chatbot can talk about, for eg:

```
('(.*) your day', ['It was good, thanks for asking!',
        'Ah, I had a long day at the office... how about
        you?']),
```

Pushing to GitHub

Pushing to GitHub

Before continuing, you must push your work to GitHub.
In the terminal:

- Commit your changes:

```
git add .
git commit -m "Finished The Simple Bots"
```

- Push to GitHub:

```
git push
```